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Contents

Part 1. Topology	5
Lecture I. Introduction	7
1. Metric spaces	7
2. Isometries and homeomorphisms	9
3. Open sets	10
4. Topological spaces	11
Lecture II. Getting used to topological spaces	13
1. Examples of topological spaces	13
2. Bases and sub bases	14
Lecture III. Fine, coarse and in between	17
1. The discrete and indiscrete topologies	17
2. Finer and coarser	17
3. Zorn's Lemma	19
Lecture IV. Subspaces and product spaces	21
1. Subspaces	21
2. Product spaces	22
3. Postscript	24
3.1. Finite products	24
3.2. Infinite products	24
3.3. The Cantor Set	25
3.4. Cantor's ternary set	25
3.5. The devil's staircase	26
Part 2. The fundamental group	29
Lecture XIII. Category theory	31
1. Categories	31
2. Functors and natural transformations	32
3. Isomorphisms and equivalences	35
4. Universal properties	35
5. Products and other universal constructions	37
Lecture XIV. Paths and the fundamental groupoid	41
1. Motivation	41
2. Path homotopy	42
3. The fundamental groupoid	44
3.1. The path product	45

Lecture XV. The fundamental group	49
Lecture XVI. Covering Spaces	51
Bibliography	53

Part 1

Topology

LECTURE I

Introduction

The notion of a “topology” is an abstract structure intended to get at the idea of “proximity”, or “nearness.” You have encountered it in calculus in the definition of “continuity.” We will back into the definition of “topology” by starting with the definition of “continuity” and slowly removing inessential structure.

1. Metric spaces

Let’s start by recalling the definition of continuity of a function $f : \mathbb{R} \rightarrow \mathbb{R}$.

Definition 1.1. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *continuous at* $x \in \mathbb{R}$ if for every $\epsilon > 0$ there exists $\delta > 0$ for which

$$|y - x| < \delta \implies |f(y) - f(x)| < \epsilon.$$

The function f is said to be *continuous* if it is continuous at all $x \in \mathbb{R}$.

There are a lot of variations you learn about this. For example, the domain and range of f only need to be subsets of $\mathbb{R}$. You also learn a generalization for functions on $\mathbb{R}^n$, and functions from $\mathbb{R}^n$ to $\mathbb{R}^m$. This already suggests that continuity of functions is really something more general than functions from one Euclidean space to another.

Examining the definition, you can see that continuity depends only on a the notion of distance, so that means you can talk about functions being continuous as soon as you have a notion of distance. In a course in analysis you isolate this structure in the notion of a *metric space*.

Definition 1.2. A *metric spaces* is a pair (X, d) consisting of a set X and a function $d : X \times X \rightarrow \mathbb{R}$ satisfying

- (1) $d(p, q) \geq 0$, and
- (2) $d(p, q) = 0$ if and only if $p = q$
- (3) $d(p, q) = d(q, p)$
- (4) given p, q, r ,

$$d(p, q) + d(q, r) \geq d(p, r).$$

The last requirement is the *triangle inequality*.

Remark 1.3. I want to dwell on the move we just made here. It’s something very mathematical. We didn’t just decide to work with objects for which we could find some natural notion of distance. We took the notion of distance as an extra piece of structure we are going to demand, and isolated some properties of distance we hope are good enough for all of our purposes. The only one of these that is not really obvious to look for is the triangle inequality. I still find it remarkable that just how far the triangle inequality lets you go in analysis.

Example 1.4. Let $X = \mathbb{R}^n$, write

$$x = (x_1, \dots, x_n)$$

$$y = (y_1, \dots, y_n)$$

and

$$d(x, y) = |x - y| = \left((x_1 - y_1)^2 + \dots + (x_n - y_n)^2 \right)^{\frac{1}{2}}.$$

This is Euclidean space with the usual metric.

Example 1.5. Again take $X = \mathbb{R}^n$ and let

$$d_{\diamond}(x, y) = |x_1 - y_1| + \dots + |x_n - y_n|.$$

One can check that $(\mathbb{R}^n, d_{\diamond})$ is a metric space. This metric is sometimes called the *taxicab* metric and sometimes the *diamond* metric.

Example 1.6. Yet again take $X = \mathbb{R}^n$ and let

$$d_{\square}(x, y) = \max\{|x_1 - y_1| + \dots + |x_n - y_n|\}.$$

One can check that $(\mathbb{R}^n, d_{\square})$ is a metric space. The metric $d_{\square}$ is the *box* metric.

One way to picture a metric is to draw a picture of the disks.

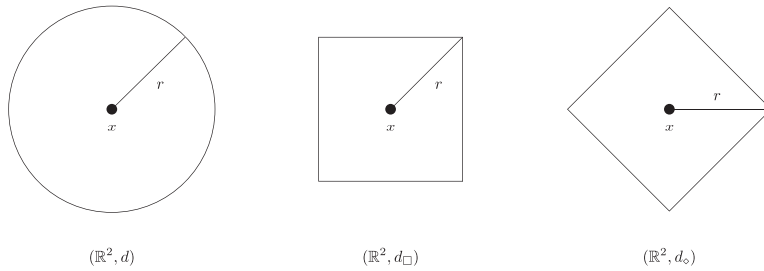
Definition 1.7. Let (X, d) be a metric space. For $r > 0$ and $x \in X$, the *closed disk* of radius r centered at x is the subset

$$D_r(x) = \{y \in X \mid d(x, y) \leq r\}$$

the *open ball* of radius r centered at x is the subset

$$B_r(x) = \{y \in X \mid d(x, y) < r\}.$$

Here are pictures of the closed disk of radius r in the three metric spaces just described



Remark 1.8. It is kind of cumbersome to use (X, d) to denote a metric space and have to keep having separate symbols for the metric d . The custom is to abuse notation and just say “let X be a metric space” and then use some on the fly symbol like d or d_X to denote the metric if you need to.

The definition of continuity works as is for metric spaces. Suppose that X and Y are metric spaces.

Definition 1.9. A function $f : X \rightarrow Y$ is *continuous* at $x \in X$ if for every $\epsilon > 0$ there is a $\delta > 0$ with the property that

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \epsilon.$$

Exercises

- 1.1.** Suppose that (X, d) is a metric space and $S \subset X$ is a subspace. Define a function $d_S : S \times S \rightarrow \mathbb{R}$ by using the metric on X : for $x, y \in S \subset X$

$$d_S(x, y) = d(x, y).$$

Show that (S, d_S) is a metric space. (In plain English, a subset of a metric space is a metric space).

- 1.2.** Suppose that X is a metric space, and $p_1, p_2, \dots$ is a sequence of points of X . A point p is a *limit* of the sequence if for every $\epsilon > 0$ there exists N so that for all $i > N$, $d(p_i, p) < \epsilon$. Show using the triangle inequality that a sequence of points in a metric space can have at most one limit.
- 1.3.** Suppose that X, Y and Z are *metric* spaces and that $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are functions. Show that if f and g are continuous then so is the composition $g \circ f : X \rightarrow Z$.

2. Isometries and homeomorphisms

Suppose that S and T are sets. Recall that a function $f : S \rightarrow T$ is

- *injective* if $f(x) = f(y) \implies x = y$
- *surjective* if for every $y \in Y$ there exists an $x \in X$ with $f(x) = y$
- *bijective* if it is both injective and surjective.

A bijective map is also called a *bijection*. There is other common terminology I will use. An *injective* map is often referred to as a *monomorphism*, a *surjective* map is often called an *epimorphism*, and *bijection* is often called an *isomorphism*. The nuances of this stuff will become more clear later in this course when we talk about categories.

A bijection $f : S \rightarrow T$ has a unique inverse function, usually denoted f^{-1} , defined by setting $f^{-1}(y)$ to be equal to the unique x for which $f(x) = y$.

Remark 2.1. Don't let this notation confuse you. The symbol f^{-1} also refers to the *preimage*. If $U \subset T$ is a subset, the *preimage* of U is the subset $f^{-1}U \subset S$ consisting of those $s \in S$ for which $f(s)$ is in U .

Definition 2.2. Suppose that X and Y are metric spaces. A function $f : X \rightarrow Y$ is an *isometry* if f is a bijection and has the property that for all $p, q \in X$,

$$d_X(p, q) = d_Y(f(p), f(q)).$$

We also say that X and Y are *isometric* if there is an isometry between them. From the point of view of metric spaces, isometric spaces are equivalent.

Example 2.3. The map “rotation by θ ” on $\mathbb{R}^2$, given by sending (x, y) to

$$\begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = (\cos(\theta)x - \sin(\theta)y, \sin(\theta)x + \cos(\theta)y)$$

is an isometry.

There is another notion of equivalence of metric spaces that is more closely related to continuity.

Definition 2.4. A function $f : X \rightarrow Y$ between metric spaces is a *homeomorphism* if f is a bijection and both f and the inverse function f^{-1} are continuous.

Example 2.5. The function

$$\begin{aligned} f : \mathbb{R}^n &\rightarrow \mathbb{R}^n \\ f(v) &= 2v \end{aligned}$$

is a homeomorphism, with inverse $v \mapsto \frac{1}{2}v$. It is *not* an isometry.

Remark 2.6. There are bijections f which are continuous but have the property that the inverse function f^{-1} is not continuous. This is a situation where the notion of bijection does not adequately capture all that we might hope it does.

We say that two metric spaces are *homeomorphic* if there is a homeomorphism between them. The notion of homeomorphism gives another kind of equivalence relation on metric spaces.

Every isometry is a homeomorphism, so isometric metric spaces are homeomorphic. However there are homeomorphic metric spaces that are not isometric.

Example 2.7. Let $X = [0, 1]$ and $Y = [0, 2]$ be the metric spaces with metric inherited as subsets of $\mathbb{R}$. The function $f(x) = 2x$ is a homeomorphism from X to Y . On the other hand X and Y are not isometric. Indeed, in X the maximum distance between any two points is 1 while in Y the distance from 0 to 2 is 2.

This fact tells us that continuity is a facet of some structure other than a metric. That is the subject of the next section.

3. Open sets

Definition 3.1. Suppose that X is a metric space. A subset $U \subset X$ is *open* if for every $x \in U$ there exists $r > 0$ for which $B_r(x) \subset U$.

The notion of open subset offers a very clean formulation of continuity.

Proposition 3.2. Suppose that X and Y are metric spaces. A function $f : X \rightarrow Y$ is continuous if and only if for every open subset $U \subset Y$ the preimage $f^{-1}U \subset X$ is open.

It will be easier to follow the proof if we restate the definition of continuity in the language of open balls. The definition of continuity at x is that given $\epsilon > 0$ there exists $\delta > 0$ with the property that

$$(3.3) \quad d_X(x, x') < \delta \implies d_Y(y, f(x')) < \epsilon.$$

Now the condition $d_X(x, x') < \delta$ is equivalent to the condition $x' \in B_\delta(x)$ and the condition $d_Y(y, f(x')) < \epsilon$ is equivalent to asking that $f(x') \in B_\epsilon(y)$. Thus the definition of continuity can be restated in terms of balls as follows:

Lemma 3.4. Suppose that X and Y are metric spaces. A map $f : X \rightarrow Y$ is continuous at $x \in X$ if and only if for every $\epsilon > 0$ there exists $\delta > 0$ with the property that

$$B_\delta(x) \subset f^{-1}B_\epsilon(f(x)).$$

□

Proof of Proposition 3.2: We will start with the forward implication. Suppose that $f : X \rightarrow Y$ is continuous and let $U \subset Y$ be open. We must show that the preimage $f^{-1}(U)$ is open. This means that for each $x \in f^{-1}(U)$ we must show that there exists $\delta > 0$ for which the open ball $B_\delta(x)$ is contained in $f^{-1}(U)$. Write $y = f(x)$. Since U is open, there is an $\epsilon > 0$ for which $B_\epsilon(y) \subset U$. Since f is continuous, there exists $\delta > 0$ with the property that $B_\delta(x) \subset f^{-1}(B_\epsilon(y))$. Since $B_\epsilon(y) \subset U$, $f^{-1}(B_\epsilon(y)) \subset f^{-1}U$ and so $B_\delta(x) \subset f^{-1}(U)$.

The reverse implication is easier. Suppose that $f^{-1}U$ is open for every open $U \subset Y$. Since $B_\epsilon(y)$ is open, $f^{-1}B_\epsilon(y)$ is open. By definition of open it contains $B_\delta(x)$ for some $\delta > 0$. $\square$

Exercises

- 3.1. Show that if $U_1, \dots, U_n$ are open subsets of a metric space X then so is $U_1 \cap \dots \cap U_n$.
- 3.2. Show that if $\{U_i\}$ is a collection of open subsets of X then the union $\bigcup_i U_i$ is open.
- 3.3. Suppose that $f : X \rightarrow Y$ is an isometry. Show that the inverse function f^{-1} is also an isometry.
- 3.4. Show that an isometry is a homeomorphism.

4. Topological spaces

We finally come to the definition central to this course.

Definition 4.1. A *topological space* is a pair $(X, \mathcal{U})$ consisting of a set X and a collection $\mathcal{U}$ of subsets of X called *open subsets*, having the following properties

- i) The empty set $\emptyset$ and X are open
- ii) A finite intersection of open sets is open
- iii) Any (possibly infinite) union of open sets is open.

Remark 4.2. It can get confusing thinking of $\mathcal{U}$ as a set whose elements are sets (which themselves have elements). Partly because of this we will try and avoid giving a name to the collection of open sets. Instead of writing “suppose that $(X, \mathcal{U})$ is a topological space,” we will just say, “suppose that X is a topological space” with the understanding that we have decided which subsets $U \subset X$ are open. Most folks find this preferable to a statement like “suppose that $(X, \mathcal{U})$ is a topological space and $U \in \mathcal{U}$ is an open subset of X .”

Suppose that X and Y are topological spaces. Motivated by Proposition 3.2 one makes the following definition.

Definition 4.3. A map $f : X \rightarrow Y$ is *continuous* if for every open $U \subset Y$ the preimage $f^{-1}(U) \subset X$ is open.

We will see one can go quite far with this very basic notion.

Exercises

- 4.1. Suppose that X , Y and Z are *topological* spaces and that $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are functions. Show that if f and g are continuous then so is the composition $g \circ f : X \rightarrow Z$. Note how much cleaner the proof is compared to Exercise [1.3](#).

LECTURE II

Getting used to topological spaces

We've now arrived at the idea of working in a topological space. To go further it will help to have some examples.

1. Examples of topological spaces

Definition 1.1. The *Sierpinski space* is the topological space $(S, \mathcal{T})$ with

$$S = \{o, c\}$$

with the collection $\mathcal{T}$ of open sets consisting of

$$\emptyset, S, \text{ and } \{o\}.$$

Up until now we've only talked about topological spaces that come from metric spaces. The Sierpinski space is an example of one that doesn't. Let's say that a topological space X *admits a compatible metric* if there is a metric d , making (X, d) into a metric space whose underlying topological space is X .

Proposition 1.2. *The Sierpinski space does not admit a compatible metric.*

Proof: Suppose that d is a metric on S and let $r = d(o, c)$. Then

$$B_r(c) = \{c\}$$

must be open. But $\{c\}$ is not open in the Sierpinski topology. $\square$

Exercises

- 1.1.** In fact the Sierpinski space gives us an example for Remark [I.2.6](#). Give the set $\{0, 1\}$ the topology coming from a metric, so that from the above, every subset is open. Show that the function

$$\begin{aligned} f : \{0, 1\} &\rightarrow S \\ 0 &\mapsto o \\ 1 &\mapsto c \end{aligned}$$

is continuous and a bijection, but the inverse function is not continuous.

- 1.2.** Suppose that (X, d) is a metric space. con

2. Bases and sub bases

It isn't always so convenient to describe a topology by explicitly specifying all of the open sets. In fact, when defining the metric topology we do so entirely in terms of open balls.

Definition 2.1. Suppose that X is a set. A *basis* (of subsets of X) is a collection $\mathcal{B}$ of subsets of X having the properties

- i) $\emptyset \in \mathcal{B}$ and $X \in \mathcal{B}$
- ii) If B_1 and B_2 are in $\mathcal{B}$ then for every $x \in B_1 \cap B_2$ there exists $B \in \mathcal{B}$ with $x \in B \subset B_1 \cap B_2$.

The language gets kind of formal sounding, and it will be helpful to employ some more colloquial expressions. For one thing, note that ii) is equivalent to saying that $B_1 \cap B_2$ is a union of elements of $\mathcal{B}$. It is also common to express this by saying that $B_1 \cap B_2$ is covered by the the sets $B \in \mathcal{B}$. It would be nice to call the elements $B \in \mathcal{B}$ *balls* to make the analogy with the presentation of the metric topology more clear, but that's not what people do. The elements of $\mathcal{B}$ are called *basis elements*.

Definition 2.2. Suppose that X is a topological space. A collection $\mathcal{B}$ of subsets $B \subseteq X$ is a *basis for the topology* of X if

- i) every $B \in \mathcal{B}$ is open
- ii) every open subset is a union of elements of $\mathcal{B}$.

Note that the second condition can be restated more formally as

- ii') if $U \subset X$ is open then for each $x \in U$ there exists $B \in \mathcal{B}$ with $x \in B \subset U$.

Every basis $\mathcal{B}$ is a basis for a topology. Suppose that $\mathcal{B}$ is a basis of sets on X . Call a set $U \subset X$ *$\mathcal{B}$ -open* if U is covered by elements of $\mathcal{B}$. Put in terms logical operators, U is $\mathcal{B}$ -open if for each $x \in U$ there exists $B \in \mathcal{B}$ with $x \in B \subset U$.

Proposition 2.3. *The collection of $\mathcal{B}$ -open subsets of X is a topology.*

Proof: Since $\emptyset, X \in \mathcal{B}$, $\emptyset$ and X are $\mathcal{B}$ -open. If $\{U_\alpha\}$ is a collection of $\mathcal{B}$ -open sets and $x \in \bigcup U_\alpha$ then $x \in U_\alpha$ for some α . Since U_α is $\mathcal{B}$ -open there exist $B \in \mathcal{B}$ with $x \in B \subset U_\alpha \subset \bigcup_\alpha U_\alpha$. Finally suppose that $\{U_1, \dots, U_n\}$ is a finite collection of $\mathcal{B}$ -open sets. If $x \in U_1 \cap \dots \cap U_n$ then for all i $x \in U_i$. Since U_i is $\mathcal{B}$ -open there exists a $B_i \in \mathcal{B}$ with $x \in B_i \subset U_i$. Since $\mathcal{B}$ is a basis, there is a $B \in \mathcal{B}$ with

$$x \in B \subset B_1 \cap \dots \cap B_n \subset U_1 \cap \dots \cap U_n$$

showing that $U_1 \cap \dots \cap U_n$ is open. □

Exercises

- 2.1.** Suppose that X and Y are topological spaces and $\mathcal{B}$ is a basis for the topology on X . Show that $f : X \rightarrow Y$ is continuous if and only if for each open $U \subset Y$ and each $x \in f^{-1}U$ there is a $B \in \mathcal{B}$ with $x \in B \subset f^{-1}U$.

- 2.2.** In contrast to Exercise 2.1 suppose instead that $\mathcal{B}$ is a basis for the topology of X . Show that a function $f : X \rightarrow Y$ is continuous if and only if for each open $U \subset Y$ the preimage $f^{-1}(U)$ is covered by elements $B \in \mathcal{B}$. To expand this out this means that for each $x \in f^{-1}(U) \subset X$ there is a $B \in \mathcal{B}$ with $x \in B \subset f^{-1}(U)$.

LECTURE III

Fine, coarse and in between

1. The discrete and indiscrete topologies

Every set X has two extreme topologies. The *discrete topology* on X is the topology $\mathcal{T}_{\text{discrete}}$ in which every subset of X is open, and the *indiscrete topology* is the topology in which the only open subsets are $\emptyset$ and X . The proof of the following is left as an exercise.

Proposition 1.1. *Suppose that A is a topological space. If X has the discrete topology then every function $f : X \rightarrow A$ is continuous. If X has the indiscrete topology then every function $g : A \rightarrow X$ is continuous.* $\square$

Exercises

1.1. Prove Proposition 1.1.

1.2. Show that the converse of Proposition 1.1 holds: If every function $f : X \rightarrow A$ is continuous then X has the discrete topology and if every function $g : A \rightarrow X$ is continuous then X has the indiscrete topology. (HINT: Look at what happens when $A = X$ with the discrete or indiscrete topology, and f and g are the identity maps.)

2. Finer and coarser

Suppose that $\mathcal{T}_1$ and $\mathcal{T}_2$ are two topologies on X .

Definition 2.1. The topology $\mathcal{T}_1$ is *finer* than $\mathcal{T}_2$ and $\mathcal{T}_2$ is *coarser* than $\mathcal{T}_1$ if $\mathcal{T}_2 \subset \mathcal{T}_1$.

In other words $\mathcal{T}_1$ is finer than $\mathcal{T}_2$ if $\mathcal{T}_1$ has more open sets than $\mathcal{T}_2$.

There are several equivalent ways of describing this relationship

Proposition 2.2. *Suppose that $\mathcal{T}_1$ and $\mathcal{T}_2$ are topologies on a set X and, for $i = 1, 2$, write X_i for the topological space $(X, \mathcal{T}_i)$. The following are equivalent*

- i) $\mathcal{T}_1$ is finer than $\mathcal{T}_2$;
- ii) The identity map on X induces a continuous map $X_1 \rightarrow X_2$.
- iii) If a function $f : A \rightarrow X$ is continuous when X is given the topology $\mathcal{T}_1$ then f is continuous when X is given the topology $\mathcal{T}_2$.

Proof: We'll start with i) $\iff$ ii). Write $f : X \rightarrow X$ for the identity map of X . The map $f : X_1 \rightarrow X_2$ is continuous if and only if for each $U \in \mathcal{T}_2$, $f^{-1}U = U$ is in $\mathcal{T}_1$, which is the assertion that $\mathcal{T}_2 \subset \mathcal{T}_1$. iii) $\implies$ ii) follows by taking

$A = X$ and f the identity map of X . Assuming ii), if $A \rightarrow X_1$ then $A \rightarrow X_1 \rightarrow X_2$ is continuous since compositions of continuous functions are continuous. $\square$

The relation $\mathcal{T}_2 \subset \mathcal{T}_1$ makes the set of topologies on a set X into a partially ordered set. The discrete topology is a maximal element and the indiscrete topology is a minimal element for this partial ordering.

Now suppose that $\mathcal{S}$ is any collection of subsets of X . Our next aim is to show that there is a coarsest (smallest) topology containing $\mathcal{S}$. Let's just think it through. Suppose that $\mathcal{T}$ is any topology with $\mathcal{S} \subset \mathcal{T}$. If $\{S_1, \dots, S_n\}$ is any finite collection of elements of $\mathcal{S}$ then $S_1 \cap \dots \cap S_n$ must be in $\mathcal{T}$. With that in mind, let's define a new collection $\mathcal{S}'$ of subsets of X by declaring U to be in $\mathcal{S}'$ if there exists a finite sequence $U_1, \dots, U_n$ of elements of $\mathcal{S}$ with

$$U = U_1 \cap \dots \cap U_n.$$

In other words we make $\mathcal{S}'$ by adding to $\mathcal{S}$ all finite intersections of elements of $\mathcal{S}$. Note that, by construction, $\mathcal{S}'$ is closed under finite intersections.

It's clear from the construction that if $\mathcal{T}$ is a topology containing $\mathcal{S}$ then $\mathcal{T}$ also contains $\mathcal{S}'$. Now if $\mathcal{T}$ contains $\mathcal{S}'$ it must also contain all unions of elements of $\mathcal{S}'$. So let's define a new collection of subsets $\mathcal{S}''$ by saying that $V \in \mathcal{S}''$ if there exists a (possibly infinite) collection $\{U_\alpha\} \subset \mathcal{S}'$ with

$$V = \bigcup_{\alpha} U_{\alpha}.$$

Proposition 2.3. *With the above notation, the collection $\mathcal{S}''$ is a topology. It is the smallest topology containing $\mathcal{S}$ in the sense that if $\mathcal{T}$ is a topology and $\mathcal{S} \subset \mathcal{T}$ then $\mathcal{S}'' \subset \mathcal{T}$.*

Proof: I'll just prove that $\mathcal{S}''$ is a topology. Since unions of unions are unions it is clear that a union of elements of $\mathcal{S}''$ is in $\mathcal{S}''$. To check that $\mathcal{S}''$ is closed under finite intersections, it suffices to check that if $U_1, U_2 \in \mathcal{S}''$ then $U_1 \cap U_2 \in \mathcal{S}''$. We can write

$$\begin{aligned} U_1 &= \bigcup_{\alpha} U_{\alpha} \\ U_2 &= \bigcup_{\beta} U_{\beta} \end{aligned}$$

with $U_{\alpha}, U_{\beta} \in \mathcal{S}'$ for all α and β . By distributivity,

$$\begin{aligned} U_1 \cap U_2 &= \left(\bigcup_{\alpha} U_{\alpha} \right) \cap \left(\bigcup_{\beta} U_{\beta} \right) \\ &= \bigcup_{\alpha, \beta} U_{\alpha} \cap U_{\beta}. \end{aligned}$$

Now each $U_{\alpha} \cap U_{\beta}$ is in $\mathcal{S}'$ since $\mathcal{S}'$ is closed under finite intersections. It follows that their union, $U_1 \cap U_2$ is in $\mathcal{S}''$. $\square$

Definition 2.4. A collection of subsets $\mathcal{S}$ is a *subbasis* for a topology $\mathcal{T}$ if $\mathcal{T}$ is the smallest topology containing $\mathcal{S}$.

Proposition 2.3 shows that every collection of subsets is a subbasis for a unique topology.

One nice thing about subbases is the following

Proposition 2.5. *Suppose that X and Y are topological spaces and $\mathcal{S}$ is a subbasis for the topology on Y . A function $f : X \rightarrow Y$ is continuous if and only if for every $U \in \mathcal{S}$, the preimage $f^{-1}U \subset X$ is open.*

The proof is left to the reader as Exercise 2.2 below.

Exercises

- 2.1. Suppose that $\{\mathcal{T}_\alpha\}$ is a collection of topologies on X . Show that $\mathcal{T} = \bigcap_\alpha \mathcal{T}_\alpha$ is a topology.
- 2.2. Prove Proposition 2.5
- 2.3. Flipping the setup as in Exercise II.2.2 Suppose that X and Y are topological and that $\mathcal{S}$ is a subbasis for the topology on X . Show that $f : X \rightarrow Y$ is continuous if and only if for each open $U \subset Y$ and each $x \in f^{-1}U$, there exist $S_1, \dots, S_n$ (for some n) with

$$x \in S_1 \cap \dots \cap S_n \subset f^{-1}U.$$

3. Zorn's Lemma

I mentioned Zorn's Lemma very briefly in class. It is covered in §11 of Munkres. In lecture I mentioned it in the context of showing that there is a *smallest* (= most coarse) topology containing a given set $\mathcal{S}$ of subsets of a set X . This doesn't really require Zorn's lemma (see Exercise 4 of §13 in Munkres), but I'll put the discussion of Zorn's Lemma here anyway, since we will need it later.

I'm going to use some notation that has become standard, and differs slightly from what Munkres uses.

Definition 3.1. A *partially ordered set* is a set X with a relation $\leq$ satisfying the following properties

- i) (Reflexive) for all $a \in X$ $a \leq a$.
- ii) (Transitive) if $a \leq b$ and $b \leq c$ then $a \leq c$.
- iii) (weakly antisymmetric) if $a \leq b$ and $b \leq a$ then $a = b$.

Now before I go further I should spell out what a *relation* is.

Definition 3.2. Let X be a set. A *relation* on X is a subset $R \subset X \times X$.

That definition is very brief, and is logically how one formulates a relation. But it isn't quite how we think of it. We think of a relation as something that two elements of X can be in. So instead of writing $(x, y) \in R$ we write something like $x \sim_R y$, and think of $x \sim_R y$ as an expression that evaluates to **TRUE** if $(x, y) \in R$ and to **FALSE** if $(x, y) \notin R$.

Example 3.3. The set $\mathbb{N} \times \mathbb{N}$ is a partially ordered set under the relation

$$(a, b) \leq (c, d)$$

if and only if

$$a \leq c \text{ and } b \leq d$$

In a partially ordered set, it might be the case that there are a and b for which neither $a \leq b$ nor $b \leq a$ holds. Such a and b are called *incomparable*.

Definition 3.4. An *ordered set* is a partially ordered set with the property that for all a, b , $a \leq b$ or $b \leq a$.

Other common names for ordered sets are *linearly ordered sets* and *totally ordered sets*.

A partial ordering on a set X induces a partial ordering on any subset $S \subset X$. If the partial ordering on X is a total ordering then so is the induced ordering on any subset $S \subset X$. It is possible for the ordering of a subset $S \subset X$ to be a total ordering even when the ordering of X is only partial.

Example 3.5. Let $X = \mathbb{N} \times \mathbb{N}$ with the partial ordering described in Example 3.3. The subset $S = \{(n, 0) \mid n \in \mathbb{N}\}$ (the x -axis) is totally ordered.

Definition 3.6. Suppose that X is a partially ordered set.

i) An element $b \in X$ is *maximal* if $a \geq b \implies a = b$ (ie, there are no elements strictly greater than b).

ii) Suppose that $S \subset X$. An element $b \in X$ is an *upper bound* of S if

$$a \in S \implies a \leq b.$$

iii) An element $b \in X$ is a *maximum* if it is an upper bound for X , ie, $a \in X \implies a \leq b$.

Now we can state Zorn's Lemma

Theorem 3.7 (Zorn's Lemma). *If X is a partially ordered set with the property that every totally ordered subset $S \subset X$ has an upper bound in X , then X has a maximal element.* $\square$

I'm not going to give a proof of Zorn's Lemma. It is equivalent to the Axiom of Choice which is independent of the Zermelo-Frankel axioms of set theory. It is customary to assume we are working in a set theory in which the Axiom of Choice holds, and hence so does Zorn's Lemma.

Remark 3.8. By reversing the equivalence relation, Zorn's Lemma also tells us that if S is a partially ordered set with the property that every totally ordered subset is bounded from below then S has a minimal element.

LECTURE IV

Subspaces and product spaces

In this lecture I discussed ways in which topologies on one space (or on a collection of spaces) induce a topology on a new space. I decided on a narrative that derived the topology from a general property it should have. I'll flip that slightly here to make the discussion more understandable.

1. Subspaces

Suppose that X is a topological space and $S \subset X$ is a subset. Let $\mathcal{T}$ be the collection of subsets $U \subset S$ for which there exists an open $V \subset X$ with $V \cap S = U$.

Lemma 1.1. *The collection $\mathcal{T}$ is a topology. This topology is called the subspace topology on S .*

Proof: I'm leaving this as an exercise (Exercise 1.1). To see that the exercise applies, note that if $f : S \rightarrow X$ denotes the inclusion then $f^{-1}U = U \cap S$. $\square$

Example 1.2. The set of rational numbers gets a topology as a subset $\mathbb{Q} \subset \mathbb{R}$.

Example 1.3. The topology on $\mathbb{Z}$ obtained by regarding $\mathbb{Z}$ as a subset of $\mathbb{R}$ is the discrete topology. Indeed

$$\{n\} = \mathbb{Z} \cap (n - \frac{1}{2}, n + \frac{1}{2}).$$

The subspace topology has an important property:

Proposition 1.4. *Suppose that X is a topological space and $S \subset X$ is a subset, regarded as a topological space with the subspace topology, and A is another topological space. A function $f : A \rightarrow S$ is continuous if and only if the function $A \rightarrow S \rightarrow X$ is continuous.* $\square$

In fact this property characterizes the subspace topology.

Proposition 1.5. *Suppose that X is a topological space and $S \subset X$ is a subset and $\mathcal{T}$ is a topology on S . Show that if $\mathcal{T}$ has the property that a map $A \rightarrow S$ is continuous if and only if $A \rightarrow S \rightarrow X$ is continuous then $\mathcal{T}$ is the subspace topology.*

Proof: I'm going to use a cool trick to prove this. Suppose $\mathcal{T}$ has the property that a map $A \rightarrow S$ is continuous if and only if $A \rightarrow S \rightarrow X$. Let S_{subspace} denote S with the subspace topology. To show that $\mathcal{T}$ is the subspace topology it suffices, by Proposition III.2.2 to show that the maps

$$\begin{aligned} S_{\text{subspace}} &\rightarrow S \\ S &\rightarrow S_{\text{subspace}} \end{aligned}$$

given by the identity map of S are continuous. Let's start with the first one. Since the composition $S_{\text{subspace}} \rightarrow S \rightarrow X$ is continuous, the property of $\mathcal{T}$ implies that the map $S_{\text{subspace}} \rightarrow S$ is continuous. Continuity of the second follows from the same argument. $\square$

Exercises

- 1.1.** Suppose that S is a set, X is a topological space and $p : S \rightarrow X$ is a function (not necessarily an inclusion).
- (a) Let $\mathcal{T}$ be the collection of subsets of S of the form $p^{-1}U$ with $U \subset X$ open. Show that $\mathcal{T}$ is a topology.
 - (b) Suppose that A is a topological space. Show that a function $f : A \rightarrow S$ is continuous if and only if the composition $p \circ f : A \rightarrow X$ is continuous.
 - (c) Prove the analogue of Proposition 1.5 for the topology on S .
- 1.2.** Suppose that X is a topological space, Y is a set, and $p : X \rightarrow Y$ is a function. Let $\mathcal{T}$ be the collection of subsets $U \subset Y$ having the property that $p^{-1}U \subset X$ is open.
- (a) Show that the collection $\mathcal{T}$ is a topology. When p is surjective, this is called the *quotient topology*.
 - (b) Show that the topology $\mathcal{T}$ has the property that $f : Y \rightarrow Z$ is continuous if and only if the composite $f \circ p : X \rightarrow Z$ is continuous.
 - (c) Show that if $\mathcal{T}'$ is a topology on Y having the property that $f : Y \rightarrow Z$ is continuous if and only if $f \circ p : X \rightarrow Z$ is continuous, then $\mathcal{T}' = \mathcal{T}$.

2. Product spaces

(There is a lot more on this in §19 of Munkres.)

Now suppose that X and Y are topological spaces. We now describe a topology on the Cartesian product $X \times Y$.

Definition 2.1. An *open box* in $X \times Y$ is a subset of the form $U \times V \subset X \times Y$ with $U \subset X$ and $V \subset Y$ open.

Example 2.2. The set

$$(a, b) \times (c, d) \subset \mathbb{R} \times \mathbb{R}$$

is an open box.

Proposition 2.3. The collection $\mathcal{B}_{\text{box}}$ of open boxes in $X \times Y$ is a basis.

Proof: By definition $\emptyset = \emptyset \times \emptyset$ and $X \times Y$ are open boxes. Suppose that $U_1 \times V_1$ and $U_2 \times V_2$ are open boxes. Then

$$(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2)$$

is also a box open subset. It follows that finite intersections of elements $\mathcal{B}_{\text{box}}$ are in $\mathcal{B}_{\text{box}}$. $\square$

Definition 2.4. The *product topology* on $X \times Y$ is the topology with basis $\mathcal{B}_{\text{box}}$.

Before turning to the characterizing property of the product topology, we single out a useful fact.

Definition 2.5. An *open cylinder set* in $X \times Y$ is a subset of the form $U \times Y$ or $X \times V$.

Proposition 2.6. *The collection $\mathcal{S}_{cyl}$ is a subbasis for the product topology.*

Proof: Note that every open cylinder set is an open box. Since

$$U \times V = (U \times Y) \cap (X \times V)$$

every open box set is a finite intersection of open cylinder sets. $\square$

Write $\pi_X : X \times Y \rightarrow X$ and $\pi_Y : X \times Y \rightarrow Y$ for the two projections, so that

$$\pi_X(x, y) = x$$

$$\pi_Y(x, y) = y.$$

Lemma 2.7. *When $X \times Y$ is given the product topology each of the projections π_X and π_Y is continuous.*

Proof: By interchanging the roles of X and Y , it suffices to show that π_X is continuous. If $U \subset X$ is open, then $\pi_X^{-1}U$ is the cylinder set $U \times Y$ and is open. $\square$

Note that Proposition 2.6 and the proof above show that the product topology is the smallest topology for which the projection maps are continuous. One can express this fact in a different way.

Proposition 2.8. *Suppose that X , Y and Z are topological spaces and give $X \times Y$ the product topology. A function $f : Z \rightarrow X \times Y$ is continuous if and only if each of the factors*

$$\begin{aligned} Z &\xrightarrow{f} X \times Y \xrightarrow{\pi_X} X \\ Z &\xrightarrow{f} X \times Y \xrightarrow{\pi_Y} Y \end{aligned}$$

is continuous.

Proof: The “only if” direction follows from the fact that a composition of continuous functions is continuous. For the other direction suppose that $\pi_X \circ f$ and $\pi_Y \circ f$ are continuous. If $U \subset X$ and $V \subset Y$ are open then by assumption

$$(\pi_X \circ f)^{-1}U = f^{-1}\pi_X^{-1}U = f^{-1}(U \times Y)$$

and

$$(\pi_Y \circ f)^{-1}V = f^{-1}\pi_Y^{-1}V = f^{-1}(X \times V)$$

are open. Since the open cylinder sets form a subbasis for the product topology, the result follows from Proposition III.2.5. $\square$

Exercises

2.1. Suppose that X is a set and $\mathcal{T}_1$ and $\mathcal{T}_2$ are two topologies on X .

- (a) Show that if for every topological space A , a function $f : A \rightarrow X$ is $\mathcal{T}_1$ -continuous then f is also $\mathcal{T}_2$ -continuous, then $\mathcal{T}_1 \subset \mathcal{T}_2$. HINT: *This fancy sounding statement is very useful, but easier to prove than you might think. What happens if you just take $A = X$ and $f : A \rightarrow X$ the identity map.*
- (b) Show that $\mathcal{T}_1 = \mathcal{T}_2$ if and only if for every topological space A a function $f : A \rightarrow X$ is $\mathcal{T}_1$ continuous if and only if f is $\mathcal{T}_2$ -continuous.
- 2.2.** Suppose that X and Y are topological spaces and $A \subset X$, $B \subset Y$ are two subsets, with the subspace topology. Show using Proposition 2.8 that the product topology on $A \times B$ is the same as the subspace topology it acquires as the subset $A \times B \subset X \times Y$. (This is Theorem 16.3 in Munkres, but I'm looking for a different proof.)

3. Postscript

3.1. Finite products. The product topology works as is for finite products. If $X_1, \dots, X_n$ are topological spaces one defines an open n -cube in $X_1 \times \dots \times X_n$ to be a subset of the form $U_1 \times \dots \times U_n$ with $U_i \subset X_i$ open sets. These form a basis for a topology on $X_1 \times \dots \times X_n$ called the product topology. The same topology can be defined inductively by writing the iterated product as

$$(X_1 \times \dots \times X_{n-1}) \times X_n.$$

For an open $U \subset X_i$ one defines the i^{th} open cylinder set as

$$X_1 \times \dots \times X_{i-1} \times U_i \times X_{i+1} \times \dots \times X_n.$$

The open cylinder sets form a subbasis for the product topology. The product topology is characterized by the property that a map

$$f = (f_1, \dots, f_n) : Z \rightarrow X_1 \times \dots \times X_n$$

is continuous if and only if each component f_i is continuous. The component f_i is the same as the composition

$$Z \xrightarrow{f} X_1 \times \dots \times X_n \xrightarrow{\pi_i} X_i$$

where π_i is projection to the i^{th} factor.

3.2. Infinite products. The story is different for infinite products. Suppose $X_1, X_2, \dots$ is an infinite sequence of topological spaces, and consider

$$\prod_{i=1}^{\infty} X_i = X_1 \times X_2 \times \dots.$$

There are two kinds of open boxes one can consider. There are subsets of the form

$$U_1 \times U_2 \times \dots = \prod_{i=1}^{\infty} U_i$$

with $U_i \subset X_i$ open. These form a basis for a topology called the *box topology*.

There are also big boxes which are subsets of the form

$$\prod_{i=1}^{\infty} U_i$$

with $U_i \subset X_i$ open and in addition, with $U_i = X_i$ for $i \gg 0$. These form a basis for a topology called the *product topology*. The cylinder sets of the form

$$\pi_i^{-1}U_i = X_1 \times \cdots \times X_{i-1} \times U_i \times X_{i+1} \times X_{i+2} \times \cdots$$

form a subbasis for the product topology, and because of that the product topology has the property (and is characterized by the property) that a function

$$f = (f_1, f_2, \dots) : Z \rightarrow \prod X_i$$

is continuous if and only if each f_i is continuous.

3.3. The Cantor Set. I now want to consider the space

$$C = \prod_{i=1}^{\infty} \{0, 1\}$$

case in which $\{0, 1\}$ has the discrete topology. The points of C are sequences

$$(\epsilon_1, \epsilon_2, \dots)$$

with each ϵ_i equal to 0 or 1. Such a point corresponds to an infinite bit string, like

$$(0, 1, 1, 0, 0, 0, 1, 0, \dots).$$

Even though each of the factors of C has the discrete topology, C itself does not. A basis of open sets look something like the set of bit strings of the form

$$(0, 1, 1, 0, 0, 0, 1, 0, 1, *, *, \dots)$$

in which the first string of, say 9 bits are fixed, and the rest can be arbitrary.

The cantor set is not discrete.

3.4. Cantor's ternary set. There is a cool embedding of the Cantor set into the closed interval $[0, 1]$. To construct it we construct a sequence

$$[0, 1] = Z_0 \supset Z_1 \supset Z_2 \supset \cdots$$

in which each Z_i is a disjoint union of 2^i closed intervals. The Cantor set is the intersection

$$C = \bigcap_i Z_i.$$

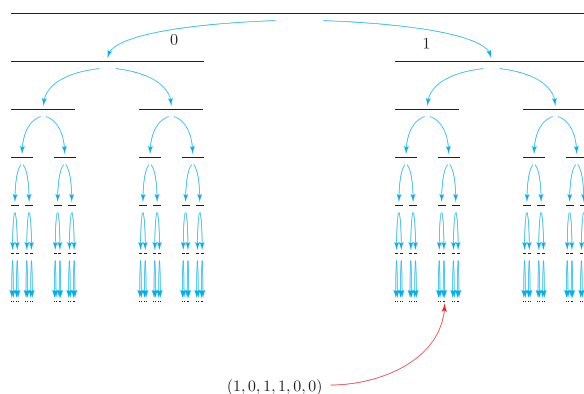
The induction starts with $Z_0 = [0, 1]$ and, having constructed Z_i one gets Z_{i+1} by removing the open middle third of each interval making up Z_i . So

$$\begin{aligned} Z_1 &= [0, 1] \setminus (1/3, 2/3) = [0, 1/3] \amalg [2/3, 1] \\ Z_2 &= ([0, 1/3] - (1/9, 2/9)) \amalg ([2/3, 1] - (7/9, 8/9)) \\ &= [0, 1/9] \amalg [2/9, 3/9] \amalg [6/9, 7/9] \amalg [8/9, 9/9] \\ Z_3 &= [0, 1/27] \amalg [2/27, 3/27] \amalg [6/27, 7/27] \amalg [8/27, 9/27] \\ &\quad \amalg [18/27, 19/27] \amalg [20/27, 21/27] \amalg [24/27, 25/27] \amalg [26/27, 27/27]. \end{aligned}$$

The formulae above aren't super enlightening. Another way to describe it is that,

$$Z_{i+1} = \frac{1}{3}Z_i \amalg \left(\frac{2}{3} + \frac{1}{3}Z_i\right).$$

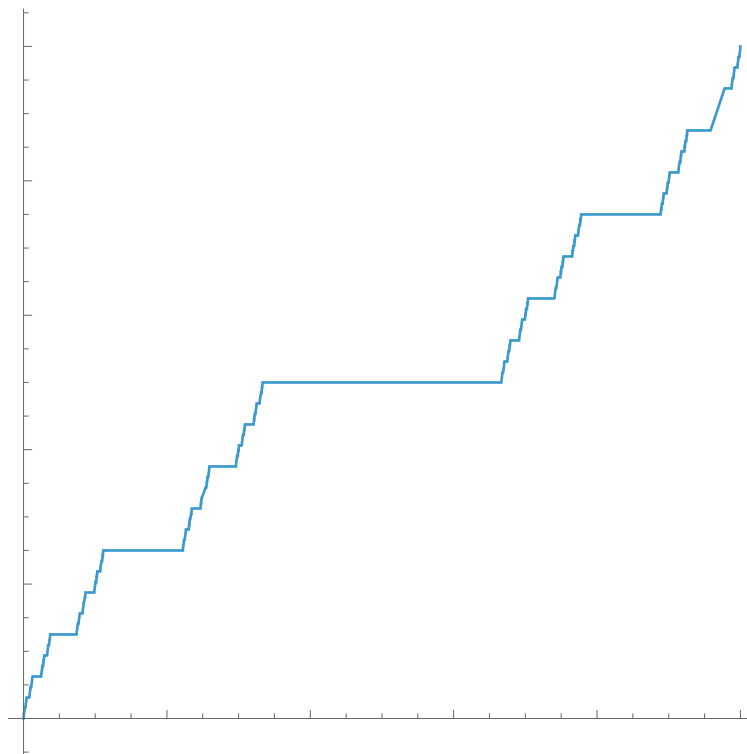
That is, having constructed Z_i you scale it by $1/3$ and take Z_{i+1} to be the union of that set and its translate by $2/3$.



The space Z_n is a disjoint union of 2^n closed intervals, each of length $1/3^n$. These intervals are naturally labeled by bit strings. You start at Z_0 and then, if the first bit is 0 you pick the interval $[0, 1/3]$ and if it is 1 you pick $[2/3, 1]$. The picture shows which interval the bit string $(1, 0, 1, 1, 0, 0)$ corresponds to. In fact this gives a bijection between the points in C with infinite bit strings and $\{0, 1\}$. You can show that this map is a homeomorphism of Cantor's ternary set with the Cantor set described in the previous section.

3.5. The devil's staircase. Under construction

In lecture I talked about the devil's staircase, which is the graph below.

**Exercises**

- 3.1.** Show that the box topology on $\prod_{i=1}^{\infty} \{0, 1\}$ is the discrete topology.

Part 2

The fundamental group

LECTURE XIII

Category theory

Categories were originally introduced by Eilenberg and MacLane with the stated purpose of providing a definition of a *natural transformation*. Over time, the idea of a category became fundamental to many fields of mathematics. It is now common practice to focus on a type of mathematical object by defining a category and working in that category. Doing so clarifies things considerably, and often highlights facets of the theory that would otherwise be difficult to perceive. An excellent reflection on Eilenberg-MacLane and the origin of categories is in [2, §7.10]. The classic book on categories is MacLane [3]. Emily Riehl's book [4] is excellent, and was based on an undergraduate class she taught at Harvard in Fall 2015 ([available here](#)).

1. Categories

Definition 1.1. A *category* $\mathcal{C}$ consists of the following data

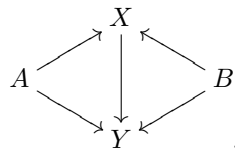
- a collection $\text{ob } \mathcal{C}$ called the *objects of* $\mathcal{C}$;
- for each $a, b \in \text{ob } \mathcal{C}$ a set $\mathcal{C}(a, b)$ called the set of *morphisms* (or *maps*) in $\mathcal{C}$ from a to b ;
- for each $a \in \text{ob } \mathcal{C}$ a morphism $\text{Id}_a \in \mathcal{C}(a, a)$ called the *identity map of* a ;
- for each $a, b, c \in \mathcal{C}$ a composition law

$$\begin{aligned} \mathcal{C}(b, c) \times \mathcal{C}(a, b) &\rightarrow \mathcal{C}(a, c) \\ (g, f) &\mapsto g \circ f. \end{aligned}$$

This data is required to satisfy the following:

- The composition law is associative: $h \circ (g \circ f) = (h \circ g) \circ f$
- The identity morphisms are units for the composition: for any a, b and any $f \in \mathcal{C}(a, b)$ one has $\text{Id}_b \circ f = f$ and $f \circ \text{Id}_a = f$.

It is customary to use some abbreviated notation. For one thing, it is clumsy to write $\text{ob } \mathcal{C}$ for the class of objects, and people usually just use $\mathcal{C}$ instead, even though the same symbol is being used to denote the category. So instead of $a \in \text{ob } \mathcal{C}$, one writes $a \in \mathcal{C}$. Also, you'll usually see $f : a \rightarrow b$ instead of $f \in \mathcal{C}(a, b)$, and relations between compositions of morphisms expressed by a commutative diagram like



Remark 1.2. The collection of objects of $\mathcal{C}$ is allowed to be a “collection” as opposed to a set. This is to allow examples like the category of all sets, and avoid the Russell paradox. The morphism sets $\mathcal{C}(a, b)$ are required to be sets. A category $\mathcal{C}$ in which $\text{ob}\mathcal{C}$ is a set is called a *small category*.

Here are some examples.

- (1) The category **Sets** is the category whose objects are sets, and in which a morphism is a function. The identity morphism is the identity map and the composition law is given by the composition of functions.
- (2) The category **Top** is the category whose objects are topological spaces with morphisms the continuous maps.
- (3) The category **Groups** is the category whose objects are groups and with morphisms the group homomorphisms.

In the above examples, categories are big things, in which a given kind of mathematical object might live. Categories can also be small things.

- (1) Given a group G one can form a category $\mathcal{B}_G$ with one object $\bullet$, morphism set $\mathcal{B}_G(\bullet, \bullet) = G$, and the composition law given by multiplication in G .
- (2) More generally given a set X and a left action of a group G on X , one can form a category X_G whose objects X , and in with $X_G(x, y) = \{g \in G \mid g \cdot x = y\}$.

We will see other examples of smaller categories below.

2. Functors and natural transformations

A map between categories is called a functor.

Definition 2.1. Suppose that $\mathcal{C}$ and $\mathcal{D}$ are two categories. A *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of

- A function

$$\begin{aligned} F : \text{ob}\mathcal{C} &\rightarrow \text{ob}\mathcal{D} \\ a &\mapsto Fa \end{aligned}$$

- For each $a, b \in \mathcal{C}$, a function

$$\begin{aligned} F : \mathcal{C}(a, b) &\rightarrow \mathcal{D}(Fa, Fb) \\ f &\mapsto Ff \end{aligned}$$

which sends identity morphisms to identity morphisms and is compatible with the composition law.

Next up is the notion of a *natural transformation*. Natural transformations take a little getting used to: they are maps between maps, and are fundamental to the theory of categories. In fact the title of the paper [1] in which Eilenberg and MacLane introduced the theory of categories is *General theory of natural equivalences*. Exercise 2.1 below might be helpful in understanding natural transformations.

Suppose that $F, G : \mathcal{C} \rightarrow \mathcal{D}$ are two functors.

Definition 2.2. A *natural transformation* from F to G , $T : F \rightarrow G$, consists of a collection of maps

$$Ta : Fa \rightarrow Ga \quad a \in \mathcal{C}$$

with the property that for each $f : a \rightarrow b$, the following diagram commutes

$$(2.3) \quad \begin{array}{ccc} Fa & \xrightarrow{Ff} & Fb \\ Ta \downarrow & & \downarrow Tb \\ Ga & \xrightarrow{Gf} & Gb. \end{array}$$

If $\mathcal{C}$ and $\mathcal{D}$ are categories then the collection of functors from $\mathcal{C}$ to $\mathcal{D}$ forms a category, in which the morphisms are natural transformations.

Here are some examples.

Example 2.4. There is a functor $U : \mathbf{Top} \rightarrow \mathbf{Sets}$ sending a topological space $(X, \mathcal{T})$ to the underlying set $U(X, \mathcal{T}) = X$.

Example 2.5. Sending a set X to the topological space $(X, \mathcal{T}_\delta)$ in which $\mathcal{T}_\delta$ is the discrete topology, gives a functor $F_\delta : \mathbf{Sets} \rightarrow \mathbf{Top}$. Similarly, sending X to $(X, \mathcal{T}_{\text{indiscrete}})$, where $\mathcal{T}_{\text{indiscrete}}$ is the indiscrete topology gives a functor $F_{\text{indiscrete}} : \mathbf{Sets} \rightarrow \mathbf{Top}$. The continuous function $(X, \mathcal{T}_{\text{indiscrete}}) \rightarrow (X, \mathcal{T}_\delta)$ given by the identity map on X defines a natural transformation $T : F_\delta \rightarrow F_{\text{indiscrete}}$.

Example 2.6. Let I be the category with two objects and one non-identity morphism as depicted below

$$\bullet \longrightarrow \bullet.$$

A functor from $F : I \rightarrow \mathcal{C}$ corresponds to a morphism $f : a \rightarrow b$ in $\mathcal{C}$. The category I is called the *free walking morphism*. Given two functors $F, G : I \rightarrow \mathcal{C}$, corresponding to two morphisms $f : a \rightarrow b$ and $g : c \rightarrow d$, a natural transformation $T : F \rightarrow G$ is a commutative diagram

$$\begin{array}{ccc} a & \xrightarrow{f} & b \\ Ta \downarrow & & \downarrow Tb \\ c & \xrightarrow{g} & d. \end{array}$$

Example 2.7. Let I be the category with three objects, and two non-identity morphisms depicted below

$$\begin{array}{ccc} & & \bullet \\ & & \downarrow \\ \bullet & \longrightarrow & \bullet \end{array}$$

A functor $I \rightarrow \mathcal{C}$ is a diagram in $\mathcal{C}$ of the form

$$\begin{array}{ccc} & & a \\ & & \downarrow \\ b & \longrightarrow & c. \end{array}$$

Example 2.8. A functor from $\mathcal{B}_G$ to the category $\mathbf{Sets}$ is a set with a left G -action.

There is also the notion of a *contravariant functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ which is like a functor, but reverses the order of composition: $F(g \circ f) = Ff \circ Fg$. Given a category $\mathcal{C}$ one can form the *opposite category* $\mathcal{C}^{\text{op}}$ with the same objects as $\mathcal{C}$ but with $\mathcal{C}^{\text{op}}(a, b) = \mathcal{C}(b, a)$ and the composition law reversed accordingly. A contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ can be identified with a (covariant) functor from $\mathcal{C}^{\text{op}}$ to $\mathcal{D}$.

Exercises

2.1. To help motivate natural transformations, let's ask a question. Is there a natural way of choosing a point s in any set S ? Let's first make something mathematically precise. We want to choose for each set S and element $s \in S$. The property that our choice be *natural* is expressed as follows: for each function $f : S \rightarrow T$ we want our choice to have the property that $f(s) = t$, that is f sends our chosen point s in S to our chosen point t in T .

- (a) Let $\text{Id} : \mathbf{Sets} \rightarrow \mathbf{Sets}$ be the identity functor with $\text{Id}(S) = S$ and $\text{Id}(f) = f$. Let $\mathbf{1} : \mathbf{Sets} \rightarrow \mathbf{Sets}$ be the functor $\mathbf{1}(S) = \{1\}$, sending every set S to the one point set whose only element is 1, and with $\mathbf{1}(f)$ the identity map of $\{1\}$. Show that the data of a natural choice of point in every set is the same thing as a natural transformation $\mathbf{1} \rightarrow \text{Id}$.
- (b) Does such a natural transformation exist? (HINT: Look at what happens when S is a one point set and T has more than one point.)

2.2. For a topology space $(X, \mathcal{T})$ let

$$\mathbf{topology}(X, \mathcal{T}) = \mathcal{T}.$$

Given a continuous function

$$f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$$

let

$$\mathbf{topology}(f) : \mathcal{T}_Y \rightarrow \mathcal{T}_X$$

be the map sending an open subset $U \subset Y$ to $f^{-1}(U) \subset X$. Show that $\mathbf{topology}$ gives a contravariant functor from **Top** to **Sets**.

2.3. This exercise concerns representable functors and the Yoneda Lemma. The Yoneda Lemma is arguably the most important theorem in category theory.

- (a) Suppose $\mathcal{C}$ is a category and $c \in \mathcal{C}$ is an object. For $x \in \mathcal{C}$, let $\mathbf{y}c(x) = \mathcal{C}(x, c)$, and for $f : x \rightarrow y$ let

$$\mathbf{y}c(f) : \mathcal{C}(y, c) \rightarrow \mathcal{C}(x, c)$$

be the function given by composition with f . Show that $\mathbf{y}c$ defines a contravariant functor from $\mathcal{C}$ to **Sets**. A functor of this form is called a *representable functor*, and $\mathbf{y}c$ is called the *functor represented by c* .

- (b) Suppose c and d are objects of $\mathcal{C}$, and $t : c \rightarrow d$ is a map. Show that composition with t defines a natural transformation $t : \mathbf{y}c \rightarrow \mathbf{y}d$.

- (c) The Yoneda Lemma says that every natural transformation $\mathbf{y}c \rightarrow \mathbf{y}d$ comes from a unique function $t : c \rightarrow d$. This part of the exercise is to prove this. Suppose that $T : \mathbf{y}c \rightarrow \mathbf{y}d$ is a natural transformation, so that for every x there is a map $T_x : \mathbf{y}c(x) = \mathcal{C}(x, c) \rightarrow \mathbf{y}d = \mathcal{C}(x, d)$ which is compatible with maps $f : x \rightarrow y$ in the sense of ???. Taking $x = c$. Then $T_c(\text{Id}_c)$ gives an element of $\mathbf{y}c(d)$ or in other words a map $t : c \rightarrow d$. Show that T is the natural transformation given by t .
- 2.4. Show that the functor **topology** from Exercise 2.2 is represented by the Sierpinski space (Definition II.1.1). (HINT: Associate to a continuous function $f : X \rightarrow S$ the preimage $f^{-1}(o)$ of the point corresponding to the open set. This gives a natural transformation from $\mathbf{y}S$ to **topology**. Show that it is a bijection for every X)

3. Isomorphisms and equivalences

Definition 3.1. A map $f : a \rightarrow b \in \mathcal{C}$ is an *isomorphism* if there exists a map $g : b \rightarrow a$ for which $g \circ f = \text{Id}_a$ and $f \circ g = \text{Id}_b$.

The map g in Definition 3.1 is called an *inverse* to f is sometimes denoted f^{-1} . Now for our first BIG THEOREM about categories.

Proposition 3.2. If $F : \mathcal{C} \rightarrow \mathcal{D}$ is a functor, and $f : a \rightarrow b$ is an isomorphism in $\mathcal{C}$, then Ff is an isomorphism.

We will use this later in the course to show that homotopy equivalent spaces have isomorphic homotopy groups.

Proof of 3.2: Having given the definition, the proof is really trivial. Fg is the inverse to Ff since $Fg \circ Ff = F(g \circ f) = F \text{Id}_a = \text{Id}_{Fa}$, etc. $\square$

So the situation is that we have useful theorem with a really trivial proof. What's the main point here? This is typical of category theory. The important idea here is the *definition* of isomorphism. We've characterized isomorphisms in terms of properties of maps. That, in some sense, is one of the main jobs of category theory: to characterize mathematical properties in terms of maps.

With the notion of "isomorphism" we've characterized certain maps in terms of properties of maps. We now turn to a simple, but useful example of characterizing a mathematical object in terms of maps.

4. Universal properties

The easiest objects to characterize are initial and terminal objects.

Definition 4.1. An *initial object* in a category $\mathcal{C}$ is an object $\emptyset \in \mathcal{C}$ with the property that there is a unique morphism from $\emptyset$ to every other object of $\mathcal{C}$. A *terminal object* is an object $*$ of $\mathcal{C}$ with the property that for each $c \in \mathcal{C}$ there is a unique morphism $c \rightarrow *$.

The empty set is an initial object in **Sets**, and any one-point set is a terminal object.

Proposition 4.2 (Uniqueness of initial and terminal objects). If $\emptyset_1$ and $\emptyset_2$ are two initial objects in $\mathcal{C}$ then there is a unique isomorphism $\emptyset_1 \rightarrow \emptyset_2$. If $*_1$ and $*_2$ are two terminal objects of $\mathcal{C}$ then there is a unique isomorphism $*_1 \rightarrow *_2$.

Before turning to the proof let's make one simplification. An initial object $\emptyset \in \mathcal{C}$ is the same thing as a terminal object in the opposite category $\mathcal{C}^{\text{op}}$. So, once we've proved Proposition 4.2 for initial objects we automatically know the result for terminal objects.

Proof of Proposition 4.2: Suppose $\emptyset_1$ and $\emptyset_2$ are two initial objects in $\mathcal{C}$. Since $\emptyset_1$ is an initial object, there is a unique map $f : \emptyset_1 \rightarrow \emptyset_2$. We will show that this map is an isomorphism. Since $\emptyset_2$ is an initial object there is a unique map $g : \emptyset_2 \rightarrow \emptyset_1$. Composing these, gives a map

$$g \circ f : \emptyset_1 \rightarrow \emptyset_1.$$

The identity map $\text{Id}_{\emptyset_1}$ is also a map from $\emptyset_1$ to $\emptyset_1$. Since $\emptyset_1$ is an initial object we must have $g \circ f = \text{Id}_{\emptyset_1}$. Similarly $f \circ g = \text{Id}_{\emptyset_2}$. $\square$

We'll see some useful instances of this result in a moment. But there's one point I want to make first. With the notion of "initial object" we've characterized an object in a category. But it's not that we've nailed down a *single* object. Instead we've identified an object which is *unique up to unique isomorphism*. That's the best you can expect in category theory, and really, it's all you should really ask for in general.

There are related notions for functors.

Definition 4.3. Suppose that $F, G : \mathcal{C} \rightarrow \mathcal{D}$ are two functors. A natural transformation $T : F \rightarrow G$ is a *natural isomorphism* if for all $c \in \mathcal{C}$ the map

$$T_c : F(c) \rightarrow G(c)$$

is an isomorphism.

Definition 4.4. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is an *equivalence of categories* if there exists a functor $G : \mathcal{D} \rightarrow \mathcal{C}$ and natural isomorphisms

$$G \circ F \rightarrow \text{Id}_{\mathcal{C}}$$

$$F \circ G \rightarrow \text{Id}_{\mathcal{D}}.$$

We thus have a hierarchy of notions of equivalence: two elements in a set can be equal, two objects in a category can be isomorphic and two categories can be equivalent.

Exercises

- 4.1. Show that if $f : a \rightarrow b$ is a map, and $g_1 : b \rightarrow a$ and $g_2 : b \rightarrow a$ are two maps satisfying

$$g_1 \circ f = g_2 \circ f = \text{Id}_a$$

$$f \circ g_1 = f \circ g_2 = \text{Id}_b$$

then $g_1 = g_2$. In the language of categories, this says that being an isomorphism is a property, and that the map g making f into an isomorphism is not extra structure. It either exists, in which case it is unique, or it doesn't.

- 4.2.** Show that a map $f : a \rightarrow b$ in $\mathcal{C}$ is an isomorphism if and only if for all $x \in \mathcal{C}$ the map $\mathcal{C}(x, a) \rightarrow \mathcal{C}(x, b)$ given by composition with f is a bijection. Put differently, an isomorphism between objects gives an identification of all of the information available from mapping into the objects. (HINT: For the only if, take $x = b$ and let $g : b \rightarrow a$ be the element of $\mathcal{C}(b, a)$ mapping to $\text{Id}_b \in \mathcal{C}(b, b)$).

5. Products and other universal constructions

We now turn to characterizations of some other objects in a category. While they may seem more general, all of them can be coerced into saying that some object is an initial or terminal object in some category. I won't spell that part out, but you'll get the idea once you start writing down proofs.

Suppose that X and Y are objects of a category $\mathcal{C}$. A *product* of X and Y is a triple (P, π_X, π_Y) consisting of an object $P \in \mathcal{C}$ equipped with maps

$$\begin{aligned}\pi_X^P : P &\rightarrow X \\ \pi_Y^P : P &\rightarrow Y\end{aligned}$$

and having the property that if $A \in \mathcal{C}$ is any object equipped with maps

$$\begin{aligned}\pi_X^A : A &\rightarrow X \\ \pi_Y^A : A &\rightarrow Y\end{aligned}$$

there is a unique map $A \rightarrow P$ making the diagram

$$\begin{array}{ccccc} & & A & & \\ & \swarrow \pi_Y^A & & \searrow \pi_X^A & \\ X & & & & Y \\ & \nwarrow \pi_Y^P & P & \nearrow \pi_X^P & \end{array}$$

commute.

Proposition 5.1 (Uniqueness of products). *Show that if $(P_1, \pi_X^{P_1}, \pi_Y^{P_1})$ and $(P_2, \pi_X^{P_2}, \pi_Y^{P_2})$ are two products of X and Y in $\mathcal{C}$ then there is a unique isomorphism $P_1 \rightarrow P_2$ making the diagram*

$$\begin{array}{ccccc} & & P_1 & & \\ & \swarrow \pi_Y^{P_1} & & \searrow \pi_X^{P_1} & \\ X & & & & Y \\ & \nwarrow \pi_Y^{P_2} & P_2 & \nearrow \pi_X^{P_2} & \end{array}$$

commute.

I'm leaving the proof as an exercise. It may feel familiar. There is a way of setting this up so that a product is a terminal object in some other category.

There is no reason to restrict to just two factors. Suppose $\mathcal{C}$ is a category, I is an indexing set, and X_α is a list of objects of $\mathcal{C}$ indexed by $\alpha \in I$.

Definition 5.2. A *product* of the X_α is an object $P \in \mathcal{C}$ equipped with maps

$$\pi_\alpha^P : P \rightarrow X_\alpha$$

having the property that if $A \in \mathcal{C}$ is another object, equipped with maps

$$\pi_\alpha^A : A \rightarrow X_\alpha$$

there is a unique map $A \rightarrow P$ which for all $\alpha \in I$ makes

$$\begin{array}{ccc} A & \xrightarrow{\quad} & P \\ \pi_\alpha^A \searrow & & \swarrow \pi_\alpha^P \\ & X_\alpha & \end{array}$$

commute.

Proposition 5.3 (Uniqueness of products). *If $(P_1, \pi_\alpha^{P_1})$ and $(P_2, \pi_\alpha^{P_2})$ are two products of the X_α in $\mathcal{C}$, then there is a unique isomorphism $P_1 \rightarrow P_2$ making the diagrams*

$$\begin{array}{ccc} P_1 & \xrightarrow{\quad} & P_2 \\ \pi_\alpha^{P_1} \searrow & & \swarrow \pi_\alpha^{P_2} \\ & X_\alpha & \end{array}$$

commute.

It's common to say *the* product of a collection X_α instead of *a* product, and denote it $\prod_\alpha X_\alpha$. Equipped with the projection mappings, it is specified uniquely up to unique isomorphism.

The general idea of characterizing an object by universal properties is to say what it means to give a map into the object. One can formulate this precisely using the Yoneda Lemma (Exercise 2.3). The information of maps into c is contained in the (contravariant) functor $\mathbf{y}c$ given by

$$\mathbf{y}c(x) = \mathcal{C}(x, c).$$

If we give a description of $\mathbf{y}c(x)$ as say, $F(x)$, for some other contravariant functor F from $\mathcal{C}$ to **Sets** then we've characterized c once we've given a natural isomorphism of $\mathbf{y}c$ with F . Indeed if we have another $c' \in \mathcal{C}$ and a natural isomorphism $\mathbf{y}c'$ with F then by composing them we get a natural isomorphism $\mathbf{y}c' \rightarrow \mathbf{y}c$ which, by the Yoneda Lemma comes from a unique isomorphism $c' \rightarrow c$. Exercise 5.4 explores this in the specific case of products.

Exercises

- 5.1. Prove Proposition 5.3.
- 5.2. Show that in **Sets** the cartesian product $\prod X_\alpha$ equipped with the projections is a product of the X_α .
- 5.3. Show that in **Top** the cartesian product $\prod X_\alpha$ in the product topology, equipped with the projections is a product of the X_α .
- 5.4. Here is another way of formulating the notion of a product. The information is exactly the same, but it emphasizes the set of maps into a product. Suppose X_α is a collection of objects of $\mathcal{C}$. A *product* of the X_α is an object $P \in \mathcal{C}$ which represents the functor $A \mapsto \prod_\alpha \mathcal{C}(A, X_\alpha)$. Show that this definition is equivalent to Definition 5.2.

- 5.5. Recall from Definition III.3.2 that a *relation* on a set X is a subset $R \subset X \times X$. The point of this exercise is to locate this, and the quotient of X by the equivalence relation generated by R in a general category. From our discussion of products we can think of R as a set, equipped with two maps to $f, g : R \rightarrow X$

$$(5.4) \quad R \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} X.$$

It makes sense to talk about this setup in any category. So let $\mathcal{C}$ be a category, and (5.4) a pair of maps in $\mathcal{C}$. A map $t : X \rightarrow Q$ is said to *coequalize* f and g if $t \circ f = t \circ g$. A map $t : X \rightarrow Q$ is a *coequalizer* of f and g if it coequalizes f and g and for any $t' : X \rightarrow Q'$ which coequalizes f and g there is a unique map $s : Q \rightarrow Q'$ for which the triangle in

$$\begin{array}{ccc} R & \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} & X \\ & \begin{array}{c} \searrow t \\ \searrow t' \end{array} & \begin{array}{c} Q \\ \downarrow s \\ Q' \end{array} \end{array}$$

commutes.

- (a) Show that if $t : X \rightarrow Q$ and $t' : X \rightarrow Q'$ are two coequalizers of f and g then the unique map $s : Q \rightarrow Q'$ making the triangle commute (ie for which $s \circ t = t'$) is an isomorphism.
- (b) Suppose that $f, g : R \rightarrow X$ are two maps in **Sets**. Let Q be the quotient of X by the equivalence relation generated by $f(r) \sim g(r)$ for all $r \in R$ (in other words, the equivalence relation generated by image of R in $X \times X$). Show that the quotient map $X \rightarrow Q$ is a coequalizer of f and g .
- 5.6. Suppose that X is a topological space and $i : A \rightarrow X$ is a map of sets (not necessarily an inclusion). Consider the following universal property: P is a topological space equipped with a map of sets $t : P \rightarrow A$ for which the composition $P \rightarrow A \rightarrow X$ is continuous. If P' is any other space, equipped with a map $t' : P' \rightarrow A$ of sets, for which $P' \rightarrow A \rightarrow X$ is continuous, then there is a unique continuous map $s : P' \rightarrow P$ making the diagram

$$\begin{array}{ccccc} P' & & \xrightarrow{\text{continuous}} & & A \\ \downarrow \exists! s & \searrow t' & & \searrow i & \\ & & A & \xrightarrow{i} & X \\ & \nearrow t & & \nearrow & \\ P & & \xrightarrow{\text{continuous}} & & A \end{array}$$

- (a) Show that if P and P' both satisfy this universal property then the unique continuous $s : P' \rightarrow P$ making the diagram commute is a homeomorphism.
- (b) Let $\mathcal{T}$ be the set of subsets of A of the form $i^{-1}(U)$ with $U \subset X$ open. Show that $\mathcal{T}$ is a topology. (NOTE: When i is an inclusion this is the subspace topology.)
- (c) Show that $P = (A, \mathcal{T})$ and $P \rightarrow A$ the identity map of underlying sets satisfies this universal property.

LECTURE XIV

Paths and the fundamental groupoid

1. Motivation

In calculus one studies *line integrals* of the form

$$\oint_{\gamma} p dx + q dy$$

along a path

$$\gamma : [0, 1] \rightarrow \mathbb{R}^2$$

from $\gamma[0] = a$ to $\gamma[1] = b$. When the vector field (technically a 1-form) is *conservative*, in the sense that

$$\frac{\partial p}{\partial y} = \frac{\partial q}{\partial x}$$

then you can find a function f with

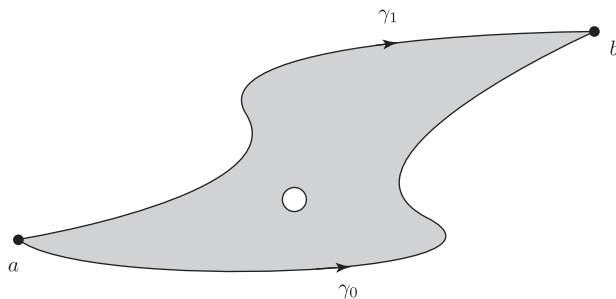
$$p = \frac{\partial f}{\partial x}$$

$$q = \frac{\partial f}{\partial y}$$

and compute

$$\oint_{\gamma} p dx + q dy = \oint_{\gamma} p dx + q dy = f(b) - f(a).$$

In particular the value of the integral is independent of the path as long as f is defined on the “area between them”. In the picture below, the gray shaded region is where f is defined, so we can’t guarantee in this case that the path integrals of γ_0 and γ_1 are the same.



Example 1.1. Let's look at

$$\oint_{\gamma} \frac{x dy - y dx}{x^2 + y^2} = \oint_{\gamma} d\theta$$

where γ is a path from $(1, 0)$ to $(-1, 0)$. For

$$\gamma(t) = (\cos(\pi t), \sin(\pi t))$$

we have

$$\begin{aligned} \oint_{\gamma} \frac{x dy - y dx}{x^2 + y^2} &= \int_0^1 \pi \frac{\cos(\pi t)^2 dt + \sin(\pi t)^2 dt}{\cos(\pi t)^2 + \sin(\pi t)^2} \\ &= \int_{t=0}^1 \pi dt = \pi. \end{aligned}$$

On the other hand, for $\gamma(t) = (\cos(-\pi t), \sin(-\pi t))$ we have

$$\begin{aligned} \oint_{\gamma} \frac{x dy - y dx}{x^2 + y^2} &= \int_0^1 -\pi \frac{\cos(-\pi t)^2 dt + \sin(-\pi t)^2 dt}{\cos(-\pi t)^2 + \sin(-\pi t)^2} \\ &= \int_{t=0}^1 -\pi dt = -\pi. \end{aligned}$$

Even though the vector field is conservative, the function f is $\theta = \arctan(y/x)$ and is not defined at the origin.

2. Path homotopy

We now want to step back and formulate the situation of the last section so that they apply to general topological spaces, and don't make use of calculus. We begin with the notion of path homotopy.

Definition 2.1. Let X be a topological space and $a, b \in X$ two points. A *path* from a to b is a continuous function

$$\gamma : [0, 1] \rightarrow X$$

with

$$\gamma(0) = a$$

$$\gamma(1) = b.$$

Definition 2.2. Suppose that γ_0 and γ_1 are two paths from a to b in X . A *path homotopy* from γ_0 to γ_1 is a continuous function

$$h : [0, 1] \times [0, 1] \rightarrow X$$

satisfying

$$h(s, 0) = \gamma_0(s)$$

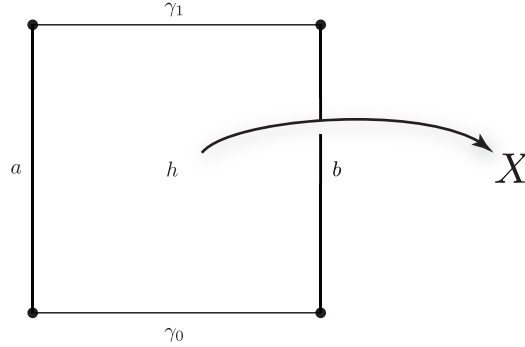
$$h(s, 1) = \gamma_1(s)$$

and for all $0 \leq t \leq 1$

$$h(0, t) = a$$

$$h(1, t) = b.$$

Here is a picture.



We will say that two paths from a to b are *path homotopic* if there is a path homotopy between them.

Proposition 2.3. *The relation “path homotopic” is an equivalence relation.*

We will deduce this from a more general result. Suppose that W is a topological space, and $A \subset W$ is a subspace and fix a continuous function $g : A \rightarrow X$. We will say that a continuous function $f : W \rightarrow X$ is an *extension of g* if for all $a \in A$, $f(a) = g(a)$.

Definition 2.4. Suppose that continuous $f_0, f_1 : W \rightarrow X$ are two extensions of g . A *homotopy from f_0 to f_1 relative to A* is a continuous function

$$h : W \times [0, 1] \rightarrow X$$

with the property that

$$h(s, 0) = f_0(s)$$

$$h(s, 1) = f_1(s)$$

and for all $0 \leq t \leq 1$ and all $a \in A$

$$h(a, t) = g(a)$$

Proposition 2.5. *With the above notation, homotopy relative to A is an equivalence relation.*

Proof: If $f : W \rightarrow X$ is an extension of g then

$$h(w, t) = f(w)$$

is a homotpy from f to f so homotopy relative to A is reflexive. For symmetry, suppose that $h : W \times [0, 1]$ is a homotopy from f_0 to f_1 . The map

$$h' : W \times [0, 1] \rightarrow X$$

$$h'(w, t) = h(w, 1 - t)$$

is then a homotopy from f_1 to f_0 . Transitivity is a little sneaky. If h_{01} is a homotopy from f_0 to f_1 and h_{12} is a homotopy from f_1 to f_2 and then the function

$$h_{02}(w, t) = \begin{cases} h_{01}(w, 2t) & 0 \leq t \leq 1/2 \\ h_{12}(w, 2t - 1) & 1/2 \leq t \leq 1. \end{cases}$$

satisfies

$$\begin{aligned} h_{02}(s, 0) &= f_0(s) \\ h_{02}(s, 1) &= f_2(s) \\ h(a, t) &= g(a). \quad a \in A \end{aligned}$$

So we are done if we know that h_{02} is continuous. This follows from Lemma 2.6 below and the fact that h_{02} is continuous on the closed subspaces $W \times [0, 1/2]$ and $W \times [1/2, 1]$. $\square$

We have used

Lemma 2.6. *Suppose that W is a topological space written as a finite union $W = W_1 \cup \dots \cup W_n$ of closed subspaces, and $f : W \rightarrow X$ is a function of the underlying sets. If the restriction of*

$$f_i : W_i \rightarrow X$$

of f to each W_i is continuous, then f is continuous.

Proof: We will show that for every closed $S \subset X$ the preimage $f^{-1}(S)$ is closed in W . Note that

$$f^{-1}S \cap W_i = f_i^{-1}(S)$$

is closed in W_i since f_i is continuous. It is therefore closed in W since W_i is closed. Thus

$$f^{-1}(S) = \bigcup_{i=1}^n f^{-1}(S) \cap W_i$$

is a finite union of closed sets, hence closed. $\square$

Proof of Proposition 2.3: This is the special case of Proposition 2.5 with $W = [0, 1]$ and $A = \{0, 1\}$. $\square$

3. The fundamental groupoid

For a space X let $\pi_{\leq 1}X(a, b)$ denote the set of path homotopy equivalent paths from a to b . First some notation. We will write

$$\gamma_0 \equiv_p \gamma_1$$

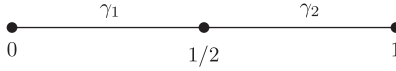
for the equivalence relation of path homotopy, and $[\gamma] \in \pi_{\leq 1}X(a, b)$ for the path homotopy equivalence class of γ .

The collection of set $\pi_{\leq 1}X(a, b)$ has some further structure.

3.1. The path product. Suppose that γ_0 is a path from a to b in X and γ_1 is a path from b to c . Imitating the proof of transitivity in Proposition 2.3 we define a new path $\gamma_0 * \gamma_1$ from a to c by

$$\gamma_0 * \gamma_1(t) = \begin{cases} \gamma_0(2t) & 0 \leq t \leq 1/2 \\ \gamma_1(2t - 1) & 1/2 \leq t \leq 1. \end{cases}$$

Here is a picture



Definition 3.1. In the situation above, the path $\gamma_0 * \gamma_1$ is called the *path product* of γ_0 and γ_1 .

The path product is not defined for every pair of paths γ_0 and γ_1 . It is only defined if $\gamma_0(1) = \gamma_1(0)$. It will be helpful to have a name for this.

Definition 3.2. Two paths γ_0 and γ_1 are *composable* if $\gamma_0(1) = \gamma_1(0)$.

Proposition 3.3. *The path product has the following properties*

i) *If γ_0 and γ_1 are composable, and*

$$\begin{aligned} \gamma_0 &\equiv_p \gamma'_0 \\ \gamma_1 &\equiv_p \gamma'_1 \end{aligned}$$

then

$$\gamma_0 * \gamma_1 \equiv_p \gamma'_0 * \gamma'_1.$$

ii) *If γ_0 and γ_1 are composable, and γ_1 and γ_2 are composable then*

$$(\gamma_0 * \gamma_1) * \gamma_2 \equiv_p \gamma_0 * (\gamma_1 * \gamma_2).$$

iii) *For a point $a \in X$ write Id_a for the constant path*

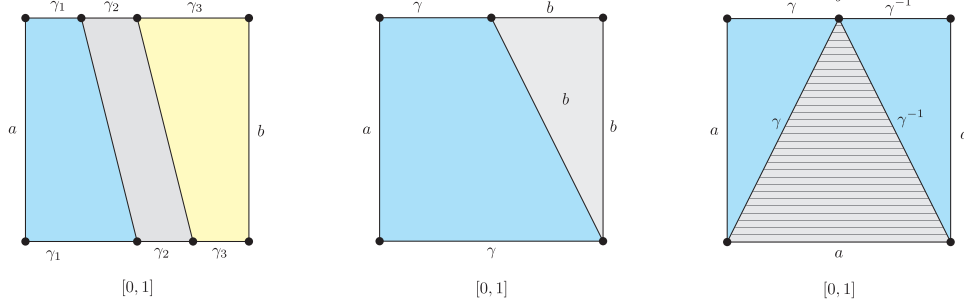
$$\text{Id}_a(t) = a.$$

*If γ is a path from a to b then $\text{Id}_a * \gamma \approx_p \gamma$ and $\gamma * \text{Id}_b \approx_p \gamma$.*

iv) *Given a path γ from a to b let γ^{-1} be the path from b to a . One has*

$$\begin{aligned} \gamma * \gamma^{-1} &\approx_p \text{Id}_a \\ \gamma^{-1} * \gamma &\approx_p \text{Id}_b. \end{aligned}$$

Proof: I'm going to leave this mostly to the reader, along with the diagrams below, indicating the homotopies.



□

One can put all of this together into a single object called the *fundamental groupoid*.

Definition 3.4. Suppose that X is a topological space. The fundamental groupoid of X is the category $\pi_{\leq 1}X$ with objects the points of X , morphisms from a to b the set $\pi_{\leq 1}X(a, b)$ of path equivalence classes of paths from a to b , identity maps given by the paths Id_a and composition given by path composition.

Remark 3.5. I swept a little technical thing under the rug. If γ_0 is a path from a to b and γ_1 is a path from b to c then the composition

$$[\gamma_1] \circ [\gamma_0] : a \rightarrow c$$

is

$$[\gamma_0 * \gamma_1].$$

Note that γ_0 and γ_1 appear in the opposite order in these two expressions. This has to do with the order in which we write function composition.

The fundamental groupoid is a category. The term *groupoid* refers to a special property it has.

Definition 3.6. A category $\mathcal{C}$ is a *groupoid* if every morphism in $\mathcal{C}$ is an isomorphism.

Exercises

3.1. Suppose that $f : X \rightarrow Y$ is a continuous function. Define

$$\pi_{\leq 1}f : \pi_{\leq 1}X \rightarrow \pi_{\leq 1}Y$$

by

$$\begin{aligned} (\pi_{\leq 1}f)(a) &= f(a) \quad \forall a \in X \\ (\pi_{\leq 1}f)(\gamma) &= f \circ \gamma \quad \forall \gamma \in \pi_{\leq 1}X(a, b) \end{aligned}$$

Show that $\pi_{\leq 1}f$ defines a functor from $\pi_{\leq 1}X$ to $\pi_{\leq 1}Y$.

- 3.2.** Suppose that $f, g : X \rightarrow Y$ are two continuous functions, and $h : X \times [0, 1] \rightarrow Y$ is a homotopy from f to g . For each $a \in X$ define $Ta \in \pi_{e1}Y(f(a), g(a))$ to be the path equivalence class of the path

$$h(a, t) : [0, 1] \rightarrow Y.$$

Show that T defines a natural transformation from $\pi_{\leq 1}f$ to $\pi_{\leq 1}g$.

Exercises

- 3.1.** Suppose that $\mathcal{C}$ is a groupoid. Show that for every $x \in \mathcal{C}$, the composition law makes $\mathcal{C}(x, x)$ into a group. Conversely, given a group G , let $\mathcal{B}_G$ be the category with one object $*$ and with $\mathcal{B}_G(*, *) = G$ as a set with a composition law. Show that $\mathcal{B}_G$ is a groupoid. In this way the notion of *groupoid* is a generalization of the notion of *group*. Sometimes *groupoids* are called *groups with many objects*.
- 3.2.** Suppose that $f : x \rightarrow y$ is an isomorphism in $\mathcal{C}$. Show that if $g_1, g_2 : y \rightarrow x$ are two inverses, then $g_1 = g_2$.
- 3.3.** Show that a map $f : x \rightarrow y$ in $\mathcal{C}$ is an isomorphism if and only if for every $z \in \mathcal{C}$ the map $f^* : \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$ is a bijection of sets.
- 3.4.** A *codiscrete groupoid* is a category in which there is a unique morphism between any two objects. From the point of view of a category theory, the best you can aspire to do when singling out an object is to isolate it as an object of a codiscrete groupoid. Show that a category $\mathcal{C}$ is a codiscrete groupoid if and only if $\mathcal{C}$ is equivalent to the “one point” category, which has a single object $*$, with $\text{hom}(*, *)$ consisting only of the identity map. (Another description is $\mathcal{B}_{\{e\}}$ where $\{e\}$ is the trivial group.)
- 3.5.** Recall that a functor

$$F : \mathcal{C} \rightarrow \mathcal{D}$$

is an *equivalence* if there exists a functor

$$G : \mathcal{D} \rightarrow \mathcal{C}$$

and a pair of natural isomorphisms

$$S : G \circ F \rightarrow \text{Id}_{\mathcal{C}}$$

$$T : F \circ G \rightarrow \text{Id}_{\mathcal{D}}.$$

Suppose that $F : \mathcal{C} \rightarrow \mathcal{D}$ is a functor, and consider the category $\mathbf{Inv}_F$ of *inverses to F* with objects the triples (G, S, T)

$$G : \mathcal{D} \rightarrow \mathcal{C}$$

$$S : G \circ F \xrightarrow{\cong} \text{Id}_{\mathcal{C}}$$

$$T : F \circ G \xrightarrow{\cong} \text{Id}_{\mathcal{D}}.$$

and in which a morphism from (G, S, T) to (G', S', T') is a natural transformation $H : G \rightarrow G'$ making the diagrams

$$\begin{array}{ccc} G \circ F & \xrightarrow{H' \circ \text{Id}_F} & G' \circ F \\ & \searrow S & \swarrow S' \\ & \text{Id}_{\mathcal{C}} & \end{array} \quad \text{and} \quad \begin{array}{ccc} F \circ G & \xrightarrow{\text{Id}_F \circ H} & F \circ G' \\ & \searrow T & \swarrow T' \\ & \text{Id}_{\mathcal{D}} & \end{array}$$

commute. Show that the category $\mathbf{Inv}_F$ is either empty, or is a codiscrete groupoid.

- 3.6.** A groupoid $\mathcal{C}$ is *connected* if for each $x, y \in \mathcal{C}$ there is a map from x to y . For $x \in \mathcal{C}$ write $G = \mathcal{C}(x, x)$. There is then a functor $\mathcal{B}_G \rightarrow \mathcal{C}$ sending $*$ to x and a map g in $\mathcal{B}_G$ to the map g in $\mathcal{C}(x, x)$ corresponding to g . Show that if $\mathcal{C}$ is connected then the functor $\mathcal{B}_G \rightarrow \mathcal{C}$ is an equivalence of categories.
- 3.7.** Suppose that X is a path connected space and choose a point $x \in X$. Show that the category of local systems on X with values in $\mathcal{C}$ is equivalent to the category of objects in $\mathcal{C}$ equipped with an action of $\pi_1(X, x)$.
- 3.8.** Suppose that $F, G : \mathcal{C} \rightarrow \mathcal{D}$ are functors from a category $\mathcal{C}$ to a groupoid $\mathcal{D}$. Show that any natural transformation $T : F \rightarrow G$ is a natural isomorphism.
- 3.9.** Suppose that $f, g : X \rightarrow Y$ are continuous maps. Show that a homotopy $h : X \times [0, 1] \rightarrow Y$ from f to g gives a natural isomorphism from $\pi_{\leq 1} f$ to $\pi_{\leq 1} g$.
- 3.10.** A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ provides, for every $x, y \in \mathcal{C}$, a map
- (3.7)
$$F : \mathcal{C}(x, y) \rightarrow \mathcal{D}(x, y).$$

The functor F is said to be

- *full* if for every $x, y \in \mathcal{C}$ the map (3.7) is an epimorphism
- *faithful* if for every $x, y \in \mathcal{C}$ the map (3.7) is a monomorphism
- *fully faithful* if for every $x, y \in \mathcal{C}$ the map (3.7) is an isomorphism.

A functor F is said to be *essentially surjective* if for every $z \in \mathcal{D}$ there exists an $x \in \mathcal{C}$ and an isomorphism of z with Fx in $\mathcal{D}$. Show that $F : \mathcal{C} \rightarrow \mathcal{D}$ is an equivalence of categories if and only if it is fully faithful and essentially surjective. (This is the analogue for categories of the fact that a map of sets is an isomorphism if and only if it is one to one and onto.)

LECTURE XV

The fundamental group

LECTURE XVI

Covering Spaces

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